السلام عليكم ورحمة الله وبركاته.. وبعد،

بين يديك ملخص شامل ومنظم لمادة CPIS-334 ، تم إعداده بعناية ليكون عوناً لك في فهم المادة .وقبل البدء، أضع بين يديك ميثاقاً للأمانة العلمية والتعاون:

⚠️ ميثاق الأمانة والمسؤولية:

هذا العمل هو جهد طالب، أضعه بين يديك بنية النفع والفائدة؛ لذا:

- لا أسامح ولا أحلل استخدامه في الغش أو أي عمل ينافي الأخلاق الإسلامية.
- يُمنع منعاً باتاً تداوله كمصدر تجاري للباعة أو استخدامه في أدوات الذكاء الاصطناعي لغرض التحايل او الغش.
- الأمانة العلمية تقتضي منك استخدامه كمرجع للفهم والمراجعة، لا كبديل عن الضمير والاجتهاد الشخصي.

🌐 خدمات إضافية عبر الموقع:

يمكنك تعزيز مذاكرتك من خلال الوصول إلى أسئلة تفاعلية تجريبية للمنهج عبر الرابط: 🔗 kau-one.web.app/quiz

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📣 التواصل معي:

الكمال لله وحده، لذا يسعدني استقبال ملاحظاتكم، تعديلاتكم، أو إضافاتكم عبر:

- الرقم المباشر: 0530717788
- الموقع الشامل: https://kau-one.web.app/contact/

توجيه إيماني:

قبل أن تبدأ، استعذ بالله من الشيطان الرجيم، وجدد نيتك بأن يكون هذا الوقت في سبيل طلب العلم النافع الذي يرفع قدرك في الدنيا والآخرة. قال رسول الله ﷺ: "مَنْ سَلَكَ طَرِيقًا يَلْتَمِسُ فِيهِ عِلْمًا سَهَّلَ اللَّهُ لَهُ بِهِ طَرِيقًا إِلَى الْجَنَّةِ" (رواه مسلم).

إعداد الطالب: إلياس حسان بابور الدفعة: 24 — تم اخذ الماده و عمل الملخص على السلايدات سنه م2026

1. What Is a Project?

- **Definition:** A project is *"a temporary endeavor undertaken to create a unique product, service, or result"*
- **Key Attributes of a Project:**
    1. **Unique Product, Service, or Result:** It has not been done in the exact same way before.
    2. **Temporary:** It has a definitive beginning and a definitive end.
        - *Reasons for ending:* Objectives are met, OR objectives cannot be met / the need no longer exists.
        - *Note:* "Temporary" does not mean short-lived (e.g., building an airport takes years). While the project itself is finite, its output usually lasts much longer.
    3. **Progressive Elaboration:** Projects develop in incremental steps. They start broadly defined; details become clearer over time through learning and updates.
    4. **Requires Resources:** Needs people, capital, tools, equipment, hardware/software, and administration.
    5. **Has a Primary Sponsor/Customer:** Provides direction and funding.
    6. **Involves Uncertainty:** Uniqueness makes it hard to pinpoint scope, schedule, or exact cost. External factors (supplier failure, sick team members) add risk.

Practical Examples

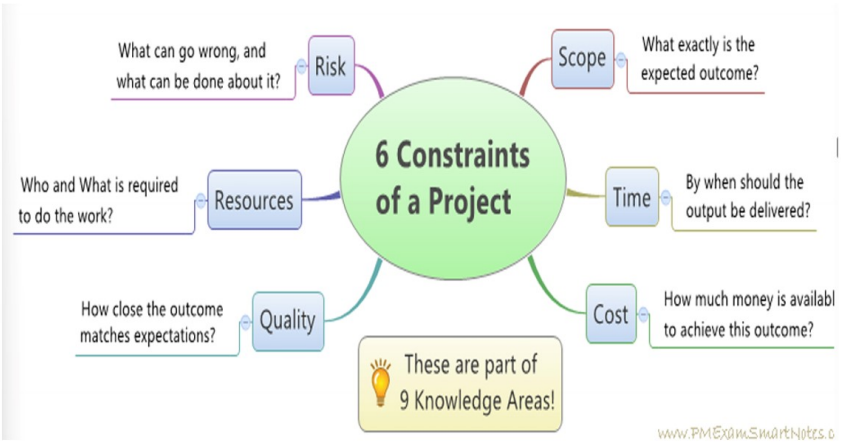
Type	**IT/Computing Example**	
Product (منتج)	Building a new smartphone or software application	بناء حاسوب جديد أو تطوير تطبيق
Service (خدمة)	Setting up free Wi-Fi infrastructure across a campus	توفير شبكة وايفاي مجانية للعملاء أو الحرم الجامعي
Result (نتيجة)	Evaluating user experience (UX) of a corporate website	إجراء دراسة لتقييم تجربة المستخدم لموقع إلكتروني

Note **: Project comes in all shapes and sizes.**

2. Project Constraints

A **constraint** is any restriction or blocker that limits a project team's options or creates trouble in reaching goals.

- **The Triple Constraint:** Scope, Time, and Cost. Project managers must continuously balance these competing goals.
- **The Quadruple Constraint:** Scope, Time, Cost + **Quality**.
- **Additional Constraints:** Risk and Resources (which directly affect the ability to balance scope, time, and cost).



What might be wrong?	Constraint Type
I miss buying the match required to light the stove.	Resources
Instead of 3/4th of a cup of water, I add 1 and a 1/2 cups.	Quality
I ended up over boiling the tea mix, and now my tea does not have the flavor I expected.	Quality
If not careful I might spill hot tea on myself while straining.	Risk
I end up spending all money on buying best quality tea powder and not have money to buy sugar.	Cost
I start making a cup of tea and add coffee powder instead of tea powder. And end up making something that resembles coffee.	Scope
I planned that it should take 4-5 minutes to make a cup of tea. But I end up taking 15 minutes or more (forgetting to fire up the stove, or keeping the flame on low).	Time

The Client Choice Rule (Reality of Life)

*Clients can pick **only two** out of three parameters:*

- ***Good + Fast** = Expensive*
- ***Good + Cheap** = Slow*
- ***Fast + Cheap** = Inferior Quality*

3. What is a Program, and Portfolio

1. **Program:** A group of related projects managed in a coordinated way to obtain benefits and control not available from managing them individually.
    a. • A program manager provides leadership and direction for the project managers heading the projects within the program + often review meetings with all their project + Many program manager worked as a project managers earlier.

2. **Portfolio:** A collection of projects, programs, and other work grouped together to achieve **strategic business goals**. They do *not* need to be directly related.
    a. portfolio managers may or may not have previous experience as project or program managers + he Help organizations make wise investment decisions decisions by helping to select and analyze projects from a strategic perspective

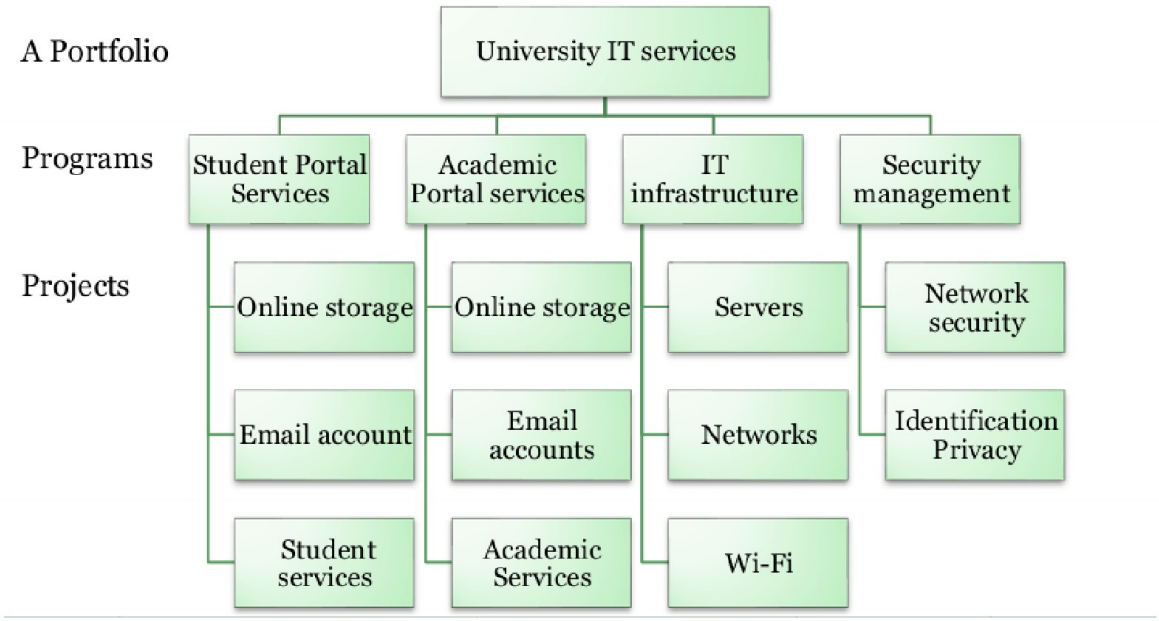
Tactical vs. Strategic Focus

- **Project Management = Doing things right** (Tactical execution: completing tasks efficiently within constraints).
- **Portfolio Management = Doing the right things** (Strategic alignment: choosing the right investments for the organization).

Core Focus		**Concept**	**Term**
التكتيك (Tactical): أداء الأعمال بطريقة صحيحة (*Doing things right*).	عمل منفرد لإنتاج شيء محدد	مشروع	**Project**
التنسيق (Coordinated Benefits): الربط بين مشاريع ذات صلة.	مجموعة مشاريع مترابطة تُدار معاً لتحقيق منافع مشتركة	برنامج	**Program**
الاستراتيجية (Strategic): اختيار الأفكار والمشاريع الصحيحة (*Doing the right things*).	مجموعة برامج ومشاريع (قد لا تكون مترابطة) تخدم أهداف المنظمة العليا	محفظة استثمارية	**Portfolio**

اللهم انفعني بما علمتني، وعلمني ما ينفعني، وزدني علماً.

University IT Hierarchy Example



4. What Is Project Management & Its Framework?

- **Definition:** The application of knowledge, skills, tools, and techniques to project activities to meet project requirements while managing constraints.
- **Stakeholders:** Anyone actively involved in or affected (positively or negatively) by the project execution or completion.
    - *Key Stakeholders:* Sponsor, Project Manager, Project Team, Support/Admin Staff, Customers, Users, Suppliers/Vendors, and **Opponents** (those whose interests are negatively impacted).

10 Project Management Knowledge Areas

1. **Integration:** Overarching function that ties all other areas together.
2. **Scope:** Defining and controlling what is and is not included in the project.
3. **Time (Schedule):** Estimating task durations, creating schedules, meeting deadlines.
4. **Cost:** Budget preparation and financial management.
5. **Quality:** Ensuring the deliverables meet required standards/needs.
6. **Human Resources (Resource):** Effectively assigning and managing team members.
7. **Communication:** Generating, distributing, and storing project information.
8. **Risk:** Identifying, analyzing, and responding to potential uncertainties.
9. **Procurement:** Acquiring external goods, tools, or vendor services.
10. **Stakeholder:** Identifying stakeholders and managing their expectations.

Tools and Techniques Examples

Knowledge Area	Tools & Techniques	
Scope	Work Breakdown Structure (WBS)	تقسيم المشروع لمهمات أصغر
Time	Gantt Charts, Network Diagrams	جدولة المهام والوقائع الزمنية
Cost	Cost Estimates, Earned Value	تقدير وإدارة الميزانية
Communication	Status & Progress Reports	تقارير حالة العمل والمستجدات
Integration	Project Management Software	برنامج شامل لإدارة العمل والربط

5. Project Success vs. Failure (النجاح والفشل)

Criteria for Success

- Met scope, time, and cost goals.
- Satisfied the customer/sponsor.
- Results achieved the primary objective (e.g., increased revenue, high ROI, solved an operational issue).

Advantages of Structured Project Management

- Better resource control (financial, human, physical).
- Improved customer relations & team morale.
- Shorter development cycles & lower costs.
- Higher profit margins & better quality.
- better internal coordination & Improve productivity

Key Reasons Projects Fail

- Poor project/program discipline.
- Lack of executive leadership support.
- No linkage to organization strategy.
- Selecting the wrong team members.
- Absence of clear success metrics or risk management.
- Inability to manage changing scope/requirements.

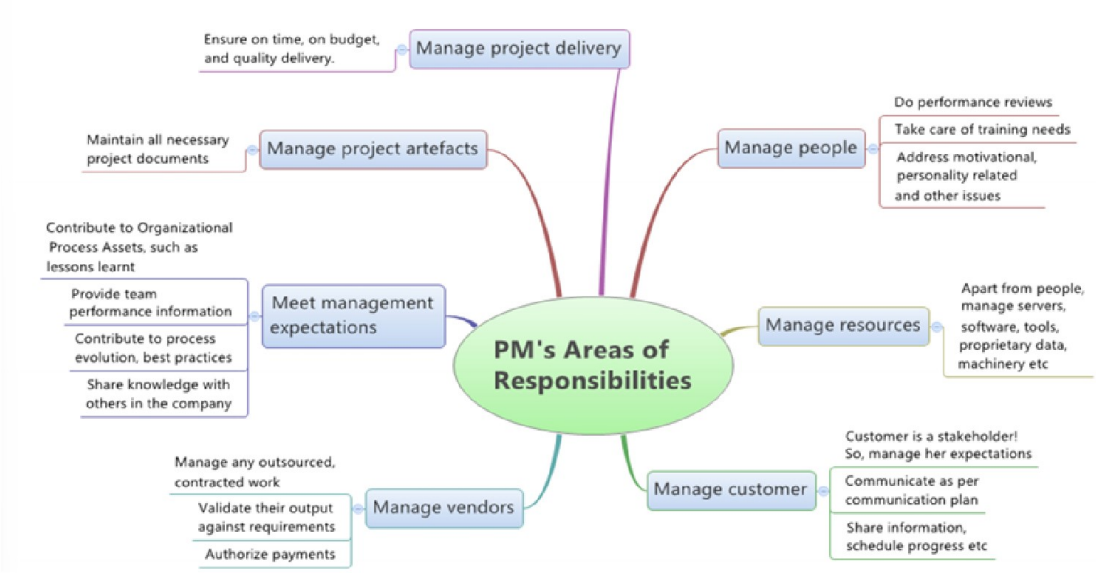
باختصار :

- **معايير النجاح:** تحقيق الأهداف ضمن الوقت والميزانية المحددَين + رضا العميل + تحقيق العائد الاستثماري المتوقع.
- **أسباب الفشل:** غياب الانضباط، ضعف دعم الإدارة العليا، عدم ربط المشروع باستراتيجية الشركة، اختيار فريق غير مناسب، إهمال إدارة المخاطر، والعجز عن التعامل مع التغييرات.

ربّ اجعل هذا العلم حجة لي لا عليّ، واجعله طريقاً لمرضاتك.

6. The Role, Skills, & Profession of a Project Manager

According to research, **97% of successful projects are led by experienced project managers**.



Skills required

1. **People skills**
2. **Leadership** focuses on long-term vision, big-picture goals, and inspiring people. Management deals with day-to-day operational execution.
3. **Soft / People Skills:** Listening, building trust, verbal communication, team building.
4. **Conflict Resolution:** Negotiating terms and reaching consensus under pressure.
5. **Critical Thinking:** Problem solving and balancing competing priorities.
6. **Integrity & Ethics:** Consistent moral conduct and transparent decision-making.

Professional Certifications & Standards(skip i think)

- **PMI (Project Management Institute):** International organization (founded 1969) offering the **PMP** (Project Management Professional) certification.
- **PRINCE2:** Structured project management methodology widely used globally and in Europe.

Ethics Code Principles (PMI)

1. Act in the best interest of society, public safety, and the environment.
2. Accept only tasks matching background, experience, and skills.
3. Honor commitments made.
4. Respect social norms, diverse viewpoints, and cultures.
5. Resolve conflict directly with affected persons.
6. Disclose potential or real conflicts of interest proactively.
7. Be truthful in all communications and representations.

7. Project Management Software Tools

1. **Low-End Tools:** Designed for small teams and single projects (Cost: <$200/user).
2. **Midrange Tools:** Designed for multiple projects, team collaboration, and tracking (Cost: 200–600/user; e.g., *Microsoft Project*).
3. **High-End / Enterprise Tools:** Complex multi-portfolio management (e.g., *Primavera*).
4. **Web-Based & Modern Cloud Tools:** Accessible via browsers/smartphones for real-time collaboration (e.g., *Clarizen, AtTask/Workfront, Asana, Jira*).

Summary

- A project is a temporary endeavor undertaken to create a unique product, service, or result
- Project portfolio management involves organizing and managing projects and programs as a portfolio of investments
- Project management is the application of knowledge, skills, tools, and techniques to project activities to meet project requirements
- Project managers play a key role in helping projects and organizations succeed
- A program is a group of related projects managed in a coordinated wa

Chapter 2 -3

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Section 1: Software Development Life Cycle (SDLC)

1. Project Phases & Life Cycle Fundamentals

- **Project Life Cycle:** A collection of project phases that defines:
  - What work will be performed in each phase.
  - What **deliverables** will be produced and when.
  - Who is involved in each phase.
  - How management will control and approve work produced in each phase.
- **Deliverable:** A tangible or intangible product/service produced as part of a project (e.g., training session, piece of hardware, a module of code).

Phase Dynamics (Early vs. Late Phases)

[Early Phase] ——> Low Resources | High Uncertainty/Risk | High Stakeholder Influence
[Middle Phase] ——> More Resources | Increasing Certainty | Execution Focus
[Final Phase] ——> High Cost to Change | Sponsor Approval | Completing Requirements

- **Early Phases:** Resource needs are lowest, uncertainty/risk is highest, and stakeholders have the greatest opportunity to influence the project's direction.
- **Middle Phases:** Certainty improves, but resource requirements spike significantly.
- **Final Phase:** Focuses on ensuring requirements were met and getting sponsor sign-off. **Making major changes in later phases is vastly more expensive.**

2. Standard SDLC Phases & Software Acquiring Methods

Software Development Life Cycle (SDLC): A framework describing the phases involved in developing and maintaining information systems.

Core SDLC Phases

1. **Analysis:** Requirements gathering and identifying system goals.
2. **Design:** Architectural specs, database design, and interface layouts.
3. **Implementation:** Writing actual code and configuring systems.
4. **Testing:** Verifying quality, functionality, and bug fixes.
5. **Support (Maintenance):** Ongoing operations, updates, and user assistance.

Software Acquisition Methods

1. Purchase New Software —> Analysis —> Research —> Purchase/Install (No dev) —> Support

2. Modify Existing ——> Analysis ——> Design ——> Implement ——> Test ——> Support

3. Integrate System —>Analysis —>Analyze Existing —>Design —>Implement —>Test —> Support

4. Outsourcing —>Analysis —>Contract —>Vendor Implement —>Vendor Test —> Vendor Support

اللهم ارزقني قوة الحفظ، وسرعة الفهم، وصفاء الذهن.

Predictive vs. Adaptive Development

- **Predictive Life Cycle (Traditional):** Used when scope is clearly articulated and schedule/cost can be accurately estimated up front.
- **Adaptive Life Cycle (ASD / Agile):** Used when requirements are unclear or change rapidly. Mission-driven and time-boxed (time-based cycles).

3. SDLC Models Comparison

[Waterfall] —— Linear & Rigid: Requirements must be locked in.

[Spiral] —— Iterative & Risk-Focused: Multi-loop evaluation.

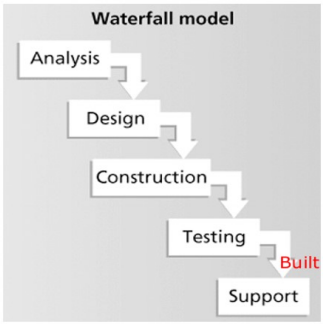
[Incremental] —— Build by Versions: Releases v1.0, v2.0, v3.0 sequentially.

[Prototyping] —— User Interface Focused: Build model -> Get feedback -> Refine.

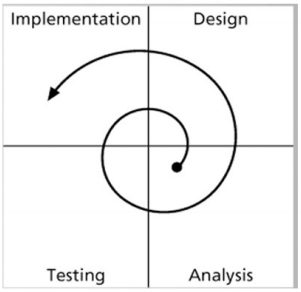
[RAD] —— High Speed / Automated: Uses CASE & JAD tools for fast code generation.

[Agile / Scrum] —— Adaptive Sprints: Short cycles producing working software rapidly.

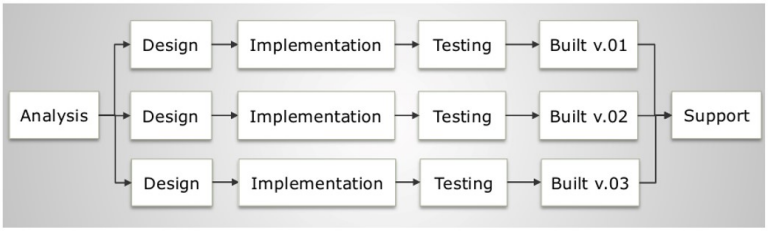
1. **Waterfall Model:**
   - Well-defined, linear, sequential stages.
   - Assumes requirements remain stable after definition.
   - Best for large-scale, high-risk systems where strict control is needed.



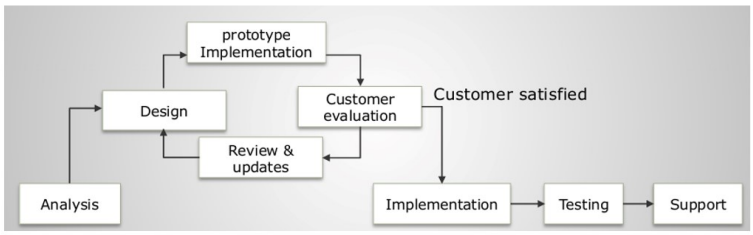
2. **Spiral Model:**
   - Iterative approach that incorporates risk analysis at each cycle.
   - The project team is open to changes and revisions later in the project life cycle - this model for projects in which changes can be incorporated with reasonable cost increases acceptable time delays



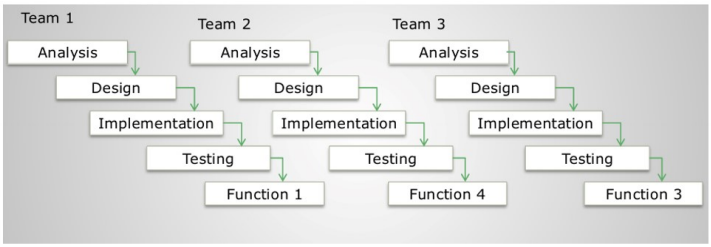
3. **Incremental Build Model:**
   - Software is delivered in operational builds/versions (e.g., Release 1.0, Build 2.0).
   - Prioritizes features based on customer urgency and resource availability (e.g., Microsoft Windows/Office updates).



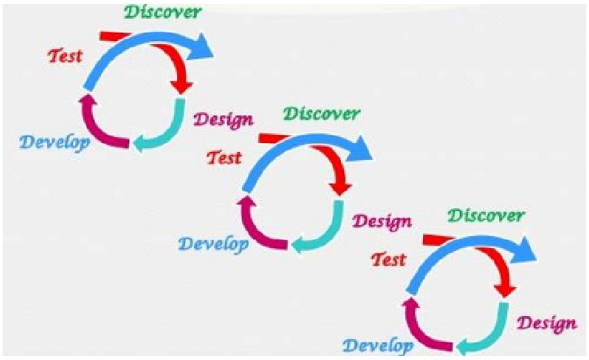
4. **Prototyping Model:**
    - Builds a working "mock-up" to help users clarify requirements.
    - Ideal for heavy UI/UX applications, mobile apps, or converting manual workflows to digital.



5. **Rapid Application Development (RAD) Model:**
    - Focuses on fast software generation using automated tools like CASE (*Computer-Aided Software Engineering*), JRP (*Joint Requirements Planning*), and JAD (*Joint Application Design*).
    - Allows multi-team parallel development on modules.



6. **Agile Software Development:**
    - Short, iterative cycles delivering working software frequently.
    - Sets time and cost boundaries, leaving **scope flexible**.



اللهم بارك في الوقت والجهد، واجعل نهاية هذا المسعى توفيقاً ونجاحاً.

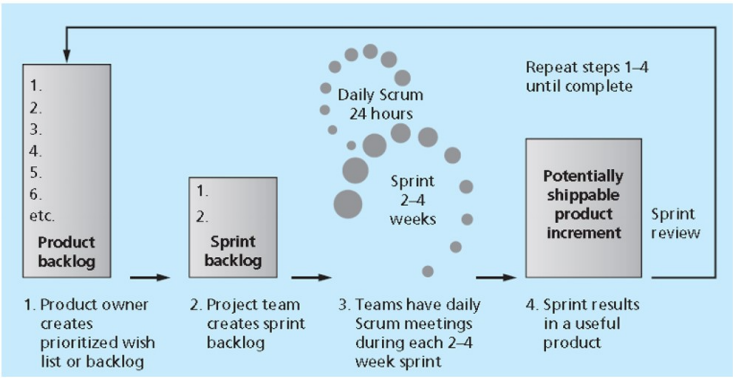
اختصار

SDLC Model	Core Characteristics	Best Used For	
Waterfall	خطي تتابعي حازم؛ التغيير فيه صعب	الأنظمة الضخمة ذات المتطلبات الثابتة والواضحة	لا ننتقل لمرحلة إلا بعد إنهاء السابقة.
Spiral	حلزوني تكراري يركز على تحليلات المخاطر	المشاريع المعقدة التي تحتاج تقييم مستمر للمخاطر	تقليل المخاطر عبر دورات متكررة.
Incremental	تقسيم النظام إلى إصدارات وتحديثات متتالية	إصدار نسخ عمل أولية ثم تحديثها (v1.0, v2.0)	إطلاق المنتج على أجزاء.
Prototyping	بناء نموذج مبدئي لأخذ ملاحظات المستخدم	مشاريع واجهات المستخدم والموقع والتطبيقات	تجربة الشاشات أولاً.
RAD	تطوير سريع جداً باستخدام أدوات توليد الكود	المشاريع ذات المواعيد الضيقة والتقارير	الاعتماد على أدوات الأتمتة.
Agile / Scrum	مرن، تكراري، يعتمد على دورات قصيرة	مشاريع التي تحتاج للتغيير السريع	تسليم برمجيات صالحة للعمل بسرعة.

4. Agile & Scrum Framework Deep Dive

- **Agile Core Principles:** Focus on collaborative effort, flexible scope, fixed timeboxes, and delivering functional software continuously.
- **Scrum Framework Components:**
  - **Product Backlog:** A prioritized list of all features and requirements desired in the final product (managed by the **Product Owner**).
  - **Sprint Backlog:** A subset of items selected from the product backlog to be completed in the upcoming sprint.
  - **Sprint:** A short time-boxed period (typically 2 to 4 weeks) where a working software increment is created note: daily scrum is 24 hour
  - **Potentially Shippable Product Increment:** The usable, tested output delivered at the end of each sprint.

Review it and repeat until the project completed



Section 2: Project Management Process Groups

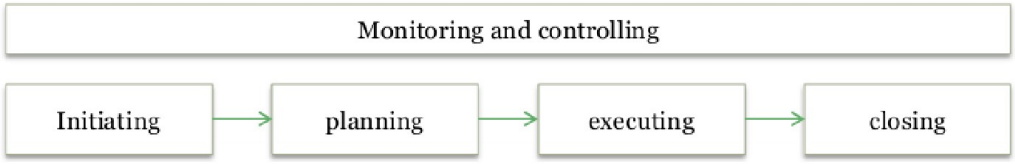
1. The 5 PM Process Groups

A **process** is a series of actions directed toward a particular result:

Inputs →{Process {Tools & Techniques}} → Outputs

The 5 Interlinked Process Groups

1. **Initiating:** Actions to define and authorize a new project or project phase.
2. **Planning:** Devising and maintaining a workable scheme to ensure the project meets its scope, time, cost, and organizational goals.
3. **Executing:** Coordinating people and resources to execute the project management plan and deliver work.
4. **Monitoring & Controlling:** Measuring progress, tracking deviations from the baseline, and executing corrective actions. (*Overlaps all other process groups throughout the project life cycle*).
5. **Closing:** Formalizing acceptance of the project/phase and bringing it to an orderly finish.



PM Process Groups & Knowledge Area Mapping (slide 21)

Knowledge Area	Project Management Process Groups				
	Initiating	Planning	Executing	Monitoring and Controlling	Closing
Project Integration Management	Develop project charter	Develop project management plan	Direct and manage project work	Monitor and control project work, Perform integrated change control	Close project or phase
Project Scope Management		Plan scope management, Collect requirements, Define scope, Create WBS		Validate scope, Control scope	
Project Time Management		Plan schedule management, Define activities, Sequence activities, Estimate activities resources, Estimate activity durations, Develop schedule		Control schedule	
Project Cost Management		Plan cost management, Estimate costs, Determine budget		Control costs	
Project Quality Management		Plan quality management	Perform quality assurance	Control quality	
Project Human Resource Management		Plan human resource management	Acquire project team, Develop project team, Manage project team		
Project Communications Management		Plan communications management	Manage communications	Control communications	
Project Risk Management		Plan risk management, Identify risks, Perform qualitative risk analysis, Perform quantitative risk analysis, Plan risk responses		Control risks	
Project Procurement Management		Plan procurement management	Conduct procurements	Control procurements	Close procurements
Project Stakeholder Management	Identify stakeholders	Plan stakeholder management	Manage stakeholder engagement	Control stakeholder engagement	

2.The Pre-Initiation Phase

- **Purpose:** Establish feasibility, justification, and key leadership before spending real project budget.
- **Key Tasks performed by Senior Management:**
    1. **Define the Triple Constraints:** Set initial rough estimates for Scope, Time, and Cost.
    2. **Identify Project Sponsor:** Find the high-level executive who will fund and champion the project.
    3. **Select the Project Manager (PM):** Choose the person responsible for day-to-day execution.
    4. **Develop the Business Case:** Create a formal document justifying why the project should happen (ROI, problem solved, strategic value).
    5. **Set Leadership Expectations:** Senior management meets with the PM to align on project governance and goals.
    6. **Determine Project Scope Boundaries:** Decide if the project is too big and should be split into smaller, manageable sub-projects.

3. Process Group 1: Initiating

This is the **official launch** of the project. Money can now be spent, and team members can be assigned.

- **Purpose:** Officially recognize and authorize a new project or project phase.
- **Main Tasks & Key Deliverables:**
    1. **Identify Stakeholders & Create Stakeholder Register:** Identify everyone impacted by the project (positively or negatively) and document their contact info, interests, and influence.
    2. **Develop Project Charter:** A short, formal document signed by the sponsor that grants the PM authority to use organizational resources.
    3. **Hold a Kick-off Meeting:** The first official meeting where stakeholders meet, align on goals, review the charter, and outline next steps.

Note : Charters are normally short and include key project information and stakeholder signatures

يا حي يا قيوم برحمتك أستغيث، أصلح لي شأني كله ولا تكلني إلى نفسي طرفة عين.

4. Process Group 2: Planning

The primary purpose of planning is to **guide execution**. If you fail to plan, you plan to fail.

Planning Outputs
├— Team Contract (Ground Rules)
├— Scope Statement (Detailed Requirements)
├— WBS (Work Breakdown Structure)
├— Gantt Chart (Schedule & Resources)
└— Risk Register (Prioritized Risks & Mitigations)

5. Process Group 3: Executing

This is where the actual work happens. It **consumes the vast majority of time and resources**.

- **Purpose:** Project managers must use their leadership skills to handle the many challenges that occur during project execution
    - Many project sponsors and customers focus on deliverables related to providing the products, services, or results desired from the project

6. Process Group 4: Monitoring & Controlling

This process group **runs concurrently with all other phases** from day one until closure.

- **Purpose:** Measure actual performance against planned performance, detect variances, and implement corrective actions.
- **Key Characteristics:**
    - Active during **all phases** of the project life cycle.
    - Influences all Knowledge Areas (Scope, Time, Cost, Quality, etc.).
- **Key Outputs:**
    - **Performance & Status Reports:** Clear metrics showing if the project is behind schedule or over budget.
    - **Change Requests:** Formal requests to modify scope, timeline, or budget when deviations occur.
    - **Project Plan Updates:** Re-baselining plans based on approved changes.

7. Process Group 5: Closing

Getting final sign-offs and bringing the project or phase to an orderly end.

- **Key Activities:**
    1. Obtain formal written acceptance from sponsor and customers.
    2. Verify all deliverables are completed and handed over.
    3. Conduct **Lessons-Learned Sessions** (What went well? What went wrong?).
    4. Archive project records into **Organizational Process Assets (OPAs)** for future reference.
- *Note:* **Failed or canceled projects MUST still undergo formal closing** so the organization can document why they failed and learn from mistakes.

8. Mapping Knowledge Areas to Practical Questions

1. Integration	How do we bring all parts together cohesive system?	كيف نربط وندمج كافة عناصر المشروع معاً؟
2. Scope	What work is required—and what is excluded?	ما هي حدود العمل المطلوبة وما المستبعد؟
3. Time (Schedule)	By when will each deliverable be done?	ما هي المواعيد والجدول الزمني للإنجاز؟
4. Cost	How much will it cost and what is the budget?	كم التكلفة الإجمالية والميزانية المتاحة؟
5. Quality	Does it meet stakeholder expectations & standards?	هل المنتج يلبي معايير الجودة المطلوبة؟
6. Human Resources	Who is needed to do the work and how are they managed?	من هم الأفراد المطلوبون وكيف نقودهم؟
7. Communications	Who needs what information, and how often?	من يحتاج المعلومات؟ وكيف ومتى ترسل له؟
8. Risk	What could go wrong, and how do we prepare?	ما الذي قد يتعطل، وكيف نتعامل معه؟
9. Procurement	What needs to be purchased or contracted externally?	ما الذي سنشتريه أو نتعاقد عليه من الخارج؟
10. Stakeholder	Who is involved/affected, and how do we engage them?	من هم التأثرون بالمشروع وكيف نكسب دعمهم؟

Summary

- SDLC phases include analysis, design, implementation testing and support
- Examples of SDLC models include Waterfall model, Spiral model, Incremental build model, Prototyping model, RAD model, Agile software development
- Brief on Agile project management
- The five project management process groups are initiating, planning, executing, monitoring and controlling, and closing
- You can map the main activities of each process group to the 9 knowledge areas.

اللهم اجعل لي نصيباً من التوفيق يقر به قلبي، وتسعد به نفسي.

Chapter 4

Chapter 4

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Chapter 4

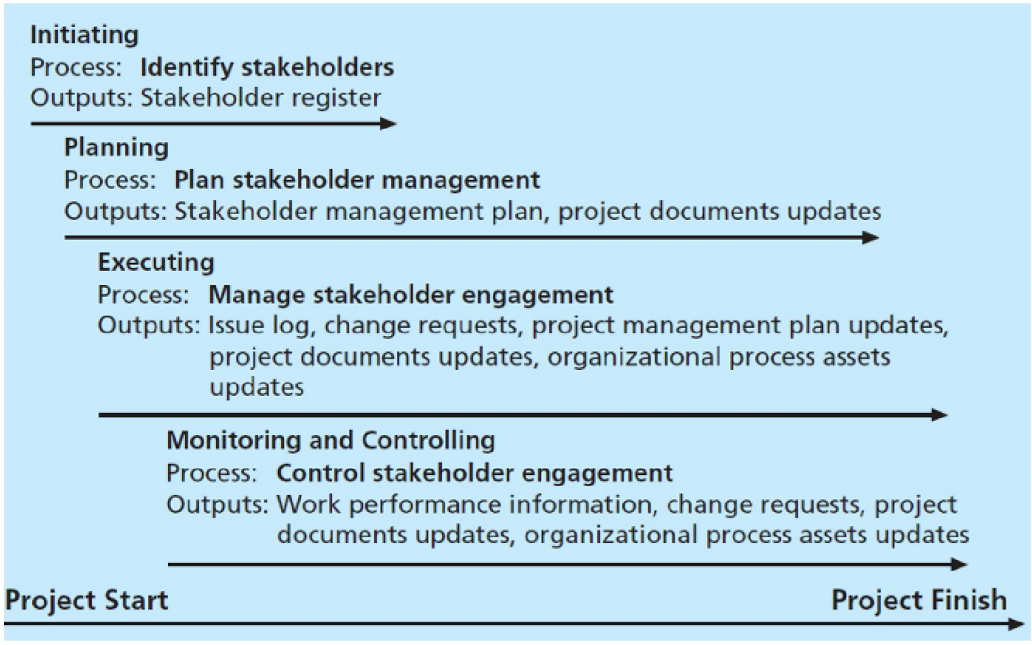
Chapter 4

1. Overview & Importance of Stakeholder Management

- **Why Stakeholders Matter:** Stakeholders are the people or entities who request projects, approve or reject them, support them, or oppose them.
- **Key Connection:** Stakeholder management borrows heavily from **Communications Management** and **Human Resource Management**.
- **Primary Goal:** Project managers (PMs) and their teams must maintain an active dialogue with stakeholders to address concerns early, resolve issues, and ensure overall satisfaction.
- **Measuring Project Success:** Success is not just meeting the triple constraint (scope, time, cost). It also includes:
    - Satisfying the customer and project sponsor.
    - Meeting quality expectations.
    - Achieving projected Return on Investment (ROI) after completion.

2. The 4 Stakeholder Management Processes Across the Lifecycle

Stakeholder management activities span across all four major project phases:



Process	Process Group / Phase	Core Action	
1. Identifying Stakeholders	Initiating	Identify everyone involved or affected and map out relationships.	حصر كافة الأطراف وتحليل أثرهم.
2. Planning Stakeholder Management	Planning	Develop strategies to effectively engage stakeholders based on their needs.	وضع استراتيجيات لإشراكهم.
3. Managing Stakeholder Engagement	Executing	Communicate, address expectations, and resolve issues dynamically.	التواصل المباشر وتلبية التوقعات.
4. Controlling Stakeholder Engagement	Monitoring & Controlling	Monitor relationships and adjust engagement strategies as needed.	مراقبة العلاقات وتعديل الاستراتيجية.

ربّ يسّر لي ما استعسرتُ من الأمور، وافتح لي أبواب فضلك وجودك.

3. Process 1: Identifying Stakeholders & Stakeholder Types

Internal vs. External Stakeholders

▼ INTERNAL STAKEHOLDERS	▼ EXTERNAL STAKEHOLDERS
• Project Sponsor	• External Customers / End Users
• Project Team	• Suppliers & Vendors
• Support / Admin Staff	• Competitors
• Top Management & Functional Managers	• Government Officials & Citizens

- **Internal Stakeholders:** Sponsor, project team, support staff, internal customers, top management, and functional managers competing for shared resources.
- **External Stakeholders:** External clients, vendors/suppliers, competitors, government regulators, and concerned community members.

The Deliverable: Stakeholder Register

The main output of the identification process is the **Stakeholder Register**. It contains three primary types of information:

1. **Identification Information:** Name, title, department, role, and contact details.
2. **Assessment Information:** Major requirements, main expectations, potential influence, and phase of highest interest.
3. **Stakeholder Classification:** Classified as internal/external, supporter/neutral/resister, or categorized by power/interest.

Stakeholder Engagement Levels

To manage stakeholders effectively, you must measure their current engagement level against the desired engagement level:

[Unaware] ⟶ [Resistant] ⟶ [Neutral] ⟶ [Supportive] ⟶ [Leading]

- **Unaware:** Completely unaware of the project and its potential impact.
- **Resistant:** Aware of the project, but opposes it due to feared negative outcomes or changes.
- **Neutral:** Aware of the project, but neither supportive nor resistant.
- **Supportive:** Aware of the project and supportive of its goals.
- **Leading:** Aware of the project and actively pushing to ensure its success.

4. Process 2: Planning Stakeholder Management

- **Purpose:** Create a formal or informal **Stakeholder Management Plan** to engage stakeholders effectively throughout the project lifecycle.
- **Key Components of the Plan:**
    1. **Current vs. Desired Engagement Levels:** Mapping where stakeholders are now versus where they need to be (e.g., moving a key manager from *Resistant* to *Supportive*).
    2. **Interrelationships:** Mapping relationships and alliances between different stakeholders.
    3. **Communication Requirements:** Determining who needs what information, in what format, and at what frequency.
    4. **Plan Maintenance:** Defining explicit methods for updating the stakeholder management plan as project dynamics change.

5. Processes 3 & 4: Managing and Controlling Stakeholder Engagement

Process 3: Managing Stakeholder Engagement

- **Action:** Direct communication and application of interpersonal and management skills.
- **Core Tasks:**
    - Active listening and open dialogue.
    - Resolving issues and conflicts before they escalate.
    - Negotiating alignment on project scope, requirements, and decisions.

Process 4: Controlling Stakeholder Engagement

- **Core Truth: "You cannot control stakeholders, but you can control their level of engagement."**
- **Action:** Continuously monitoring relationships, measuring sentiment, and tweaking engagement strategies.
- **Key Strategy:** Engagement requires an ongoing **two-way dialogue** where both sides seek clear understanding and mutually beneficial solutions.

اللهم سخر لي من الأقدار أجملها، ومن التوفيق أكمله، ومن السداد أصوبه.

Chapter 5

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Chapter 5

1. Introduction to Project Scope Management

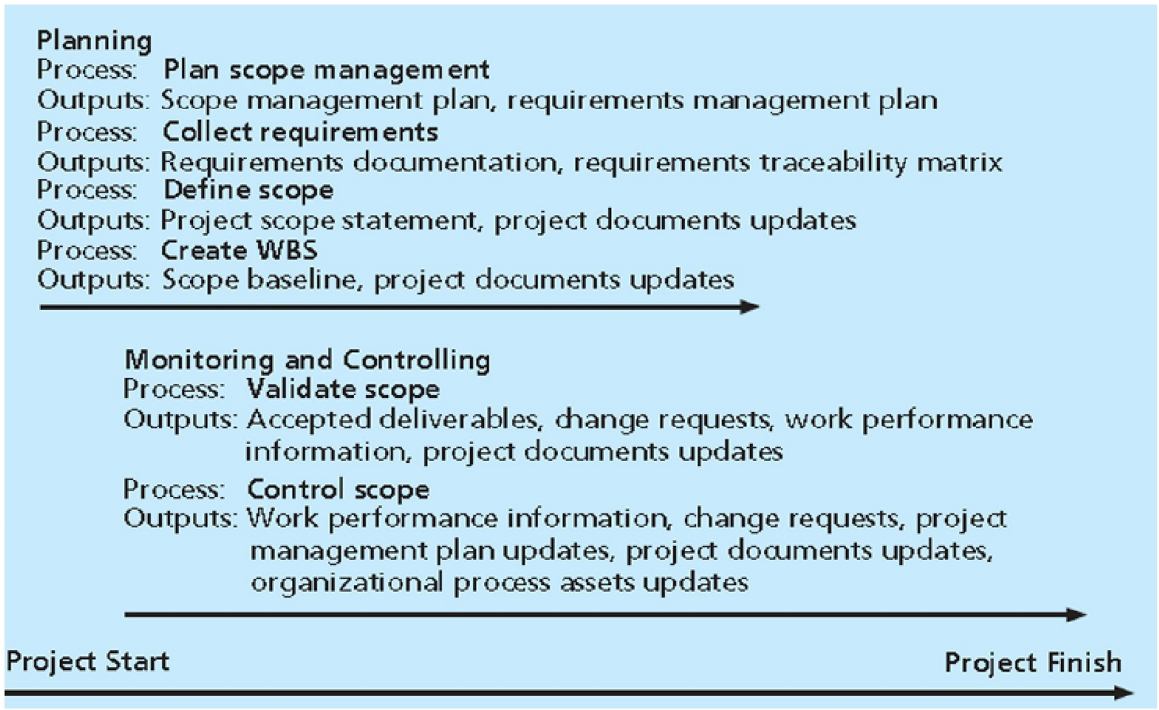
- **Scope:** All the work involved in creating the project's products or services, as well as the processes used to build them.
- **Project Scope Management:** The set of processes required to ensure the project includes **all the work required, and ONLY the work required**, to complete the project successfully.
- **Primary Objective:** Ensures that the project team and stakeholders share a common, clear understanding of:
    - What products/services the project will produce (**Product Scope**).
    - What processes will be used to build them (**Project Scope**).

2. Overview of the 6 Scope Management Processes

Project Scope Management consists of six distinct processes spanning Planning, Executing, Monitoring & Control, and Closing:

Process Group	Process	Core Deliverable / Output	
Planning	**1. Plan Scope Management**	Scope Mgmt Plan & Requirements Mgmt Plan	كيفية إعداد وإدارة النطاق.
Planning	**2. Collect Requirements**	Requirements Documentation & RTM	توثيق احتياجات أصحاب المصلحة.
Planning	**3. Define Scope**	Project Scope Statement	صياغة بيان تفصيلي لما ينطوي عليه العمل.
Planning	**4. Create WBS**	Scope Baseline (WBS, Dictionary, Scope Statement)	تفكيك المخرجات لمهام أصغر.
M&C	**5. Validate Scope**	Accepted Deliverables	القبول الرسمي للتسليمات من العميل.
M&C	**6. Control Scope**	Work Performance Info & Change Requests	إدارة وتتبع التغيرات في النطاق.

نفس الكلام الي فوق :



3. Deep Dive into the 6 Scope Processes

Process 1: Plan Scope Management

Involves determining how the project scope and requirements will be defined, validated, and managed.

- **Main Tools:** Expert Judgment and Meetings.
- **Two Key Outputs:**
    1. **Scope Management Plan:** Describes how to prepare the detailed scope statement, build and approve the WBS, obtain formal acceptance of completed deliverables, and handle scope change requests.
    2. **Requirements Management Plan:** Documents how requirements will be collected, analyzed, prioritized, tracked, and managed across iterative cycles.
        - The PMBOK describes requirements as “conditions or capabilities that must be met by the project or present in the product, service, or result to satisfy an agreement or other formally imposed specification”

Process 2: Collect Requirements

Defining and documenting the features and functions of products as well as the processes used to create them.

Requirements Elicitation Methods

- **Interviewing:** Direct Q&A with key stakeholders.
- **Brainstorming:** Group idea generation sessions.
- **Focus Groups & Facilitated Workshops:** Interactive group discussions with stakeholders/subject-matter experts.
- **Questionnaires & Surveys:** Broad data collection for large user bases.
- **Observation:** Shadowing users to see actual operational workflows.
- **Prototyping:** Building mock-ups/working models to iterate user feedback.
- **Benchmarking:** Comparing project practices or product specs against industry best practices or competitors.

Key Output: Requirements Traceability Matrix (RTM)

A grid table that links product requirements from their origin to the deliverables that satisfy them. It ensures every requirement is addressed, tested, and tracked.

Requirement No.	Name	Category	Source	Status
R32	Laptop memory	Hardware	Project charter and corporate laptop specifications	Complete. Laptops ordered meet requirement by having 16 GB of memory.

Process 3: Define Scope (تحديد النطاق)

Reviewing the project charter, requirements documents, and organizational process assets to write a detailed **Project Scope Statement**.

The Scope Statement Components

1. **Product Scope Description:** Detailed characteristics of the output.
2. **Product User Acceptance Criteria:** Clear benchmarks required for formal sign-off.
3. **Project Deliverables:** Detailed list of all physical or digital outputs.
4. **Boundaries, Constraints, & Assumptions:** Clear statements of what **IS NOT** included in the project to prevent **Scope Creep** (uncontrolled expansion of project scope without time/cost adjustments).

اللهم لا تحرمني خير ما عندك بسوء ما عندي، واكتب لي النجاح في كل خطوة.

Process 4: Create the Work Breakdown Structure (WBS)

Subdividing major project deliverables into smaller, more manageable components (**Decomposition**).

Key WBS Terminology

- **Work Breakdown Structure (WBS):** A deliverable-oriented grouping of project work defining total scope. It serves as the foundation for schedules, budgets, resource assignments, and changes.
- **Work Package:** The lowest level task entry in a WBS.
- **WBS Dictionary:** A document that explains detailed specifics for each WBS code item (task responsibility, cost estimates, start/end dates, resources needed).
- **Scope Baseline:** Formed by the **Scope Statement + WBS + WBS Dictionary**.

Approaches to Building a WBS

1. **Guidelines:** Using organizational standards or templates.
2. **Analogy Approach:** Reviewing WBS from similar past projects and customizing.
3. **Top-Down Approach:** Starting with the main project outcomes and decomposing downwards.
4. **Bottom-Up Approach:** Listing specific individual tasks first and grouping them up into deliverables.

Core Rules for Creating a WBS

- **100% Rule:** A unit of work must appear in only one place in the WBS. Sum of child items equals 100% of the parent item.
- **Single Responsibility:** Each WBS item is assigned to **only one individual**, even if many team members work on it.
- The WBS must be consistent with the way in which work is actually going to be performed; it should serve the project team first, and other purposes only if practica
- Project team members should be involved in developing the WBS to ensure consistency and buy-in
- Each WBS item must be documented to ensure accurate understanding of the scope of work included and not included in that item

Process 5: Validate Scope

Formalizing acceptance of completed deliverables through customer inspection and sign-off.

- **Key Steps:** Customers/Sponsors inspect completed deliverables → If satisfied, they sign off → If unsatisfactory, formal change requests are issued.
- **Primary Output: Accepted Deliverables**, Work Performance Information, and Change Requests.

Process 6: Control Scope

Controlling and managing changes to the project scope throughout its lifecycle.

- **Primary Mechanism: Variance Analysis** — comparing planned performance (Scope Baseline) against actual progress.
- **Key Objectives:**
    - Influence factors that cause scope changes.
    - Ensure changes are processed through the **Integrated Change Control** system.
    - Manage real adjustments when changes occur to protect time and cost baselines.
- **Primary Outputs:** Work Performance Information, Change Requests, and updates to the Project Management Plan.

4. Best Practices & Software Tools in Scope Management

Best Practices to Prevent Scope Problems

1. Realism ——> Keep projects realistic; break large projects into smaller ones.

2. User Ownership——> Involve users actively in requirements and scope verification.

3. COTS Tech ——> Use Commercial-Off-The-Shelf hardware/software before custom tech.

4. Process Focus ——> Strictly adhere to standard project management frameworks.

Software Tools Assisting Scope Management

- **Word Processors:** Crafting scope statements, requirements docs, and WBS dictionaries.
- **Spreadsheets:** Financial models, scoring matrices, and requirements traceability matrices.
- **Project Management Software (e.g., MS Project):** Creating WBS layouts and converting WBS tasks into Gantt Chart schedules.
- **Communication & Specialized Tools:** Collaboration systems to track scope change approvals.

Summary

- Project scope management includes the processes required to ensure that the project addresses all the work required, and only the work required, to complete the project successfully
- Main processes include
    - Define scope management
    - Collect requirements
    - Define scope
    - Create WBS
    - Validate scope
    - Control scope

Chapter 7

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Chapter 7

اللهم أنر بصيرتي، واشرح صدري، وثبّت في قلبي وعقلي ما ينفعني.

1. Introduction to Project Cost Management

Cost is a resource sacrificed or given up in exchange to achieve a specific objective. **Project Cost Management** includes the processes required to ensure a project is completed within an approved budget.

Why IT Projects Suffer Cost Overruns:

IT projects have a historically poor track record (overruns ranging from 27% to 180%). If managers don't focus on costs, the project risks failure, reduced quality, or customer dissatisfaction. Overruns happen because:

- **Estimates are rushed:** Done before clear system requirements exist.
- **Lack of experience:** IT managers mistakenly think estimating is an accountant's job.
- **Bias toward underestimation:** Humans naturally underestimate. Estimators forget "hidden" costs like software testing and user training.

2. Basic Financial Principles

Executive boards care about financial health, not IT jargon. IT project managers must speak their language:

- **Profits (الأرباح):** Revenues minus expenditures.
- **Profit Margin (هامش الربح):** The ratio of revenues to profits.
- **Life Cycle Costing (تكلفة دورة الحياة):** Considers the Total Cost of Ownership—meaning the cost of development *plus* long-term support and maintenance.
- **Cash Flow Analysis (تحليل التدفق النقدي):** Determines estimated annual costs and benefits to show the resulting annual cash flow.

3. Types of Costs and Benefits

- **Tangible vs. Intangible:** Tangible costs/benefits are easily measured in dollars (e.g., hardware costs). Intangible ones are difficult to measure monetarily (e.g., employee morale, customer disappointment).
- **Direct vs. Indirect:** Direct costs are exclusively for the project (e.g., salaries for the project team). Indirect costs support the project but aren't exclusive to it (e.g., office electricity).
- **Fixed vs. Variable:** Fixed costs remain constant (e.g., renting equipment for a month). Variable costs fluctuate based on conditions (e.g., number of meeting participants, material supply/demand).
- **Sunk Cost (التكلفة الغارقة):** Money already spent in the past. **Rule:** You should *not* include sunk costs when deciding whether to continue a project.
    - *Example:* Spending 50,000 SAR on a marketing study for a product that will fail. The money is gone; don't keep funding the failing project just because you already spent money on it.

4. Learning Curve Theory & Pricing

Learning Curve Theory:

States that when items are produced repetitively, the unit cost decreases in a regular pattern as more units are produced, because people and processes get faster and more efficient.

- *Hardware Example:* If a project produces 1,000 satellite handheld devices, the 1st device will cost much more to make than the 1,000th device.
- *Labor Example:* The first time an employee does a specific task, it takes longer than the tenth time they do it.

Costing vs. Pricing:

- **Cost:** The quantitative value of how much it takes to *provide* the project/service.
- **Price:** A business decision determining what the *customer pays*.
- **Formula:** Price = Cost + Profit.
- *Key Concept:* There is no simple relationship between development cost and price. Prices are influenced by broader organizational, economic, political, and business considerations.

5. Financial Reserves

Reserves are funds included in a cost estimate to mitigate risk:

- **Contingency Reserves (Known Unknowns):** Included in the baseline for predictable risks. *Example:* An organization knows it has a 20% IT staff turnover rate, so they budget extra for future recruiting and training.
- **Management Reserves (Unknown Unknowns):** Set aside for unpredictable situations. *Example:* A project manager gets sick for two weeks, or a supplier suddenly goes bankrupt.

اللهم أخرجني من ظلمات الوهم، وأكرمني بنور الفهم، وافتح عليّ بمعرفة العلم.

6. The 4 Processes of Project Cost Management

Process 1: Planning Cost Management

Determining the policies, procedures, and documentation for managing project costs. The output is a **Cost Management Plan**, which defines:

- **Level of accuracy:** Rounding guidelines (e.g., to the nearest $100) and contingency percentages.
- **Units of measure:** Each unit used in cost measurements, such as Tracking labor in hours or days.
- **Organizational procedure links:** Linking the Work Breakdown Structure (WBS) to the accounting system via Control Accounts (CA).
- **Control thresholds:** The allowed variation before action is taken (e.g., a 10% budget variance limit).
- **Rules of performance measurement:** Defining how Earned Value Management (EVM) will be tracked.
- **Reporting formats:** Frequency and format of cost reports.
- **Process descriptions:** The cost management plan would also describe how to perform all of the cost management processes

Process 2: Estimating Costs

Developing an approximation of resource costs. Labor costs often make up the largest percentage of IT budgets. Accuracy improves as time progresses.

Types of Cost Estimates:

1. **Rough Order of Magnitude (ROM):** Done very early to help make project selection decisions. Accuracy: -50% to +100%.
2. **Budgetary:** Done 1–2 years in advance to allocate budget money. Accuracy: -10% to +25%.
3. **Definitive:** Highly accurate, done before purchasing decisions. Accuracy: -5% to +10%.

Estimation Tools & Techniques:

- **Analogous (Top-Down):** Uses the actual cost of a previous, similar project. Less costly, but less accurate. Requires expert judgment and truly similar past projects.
- **Bottom-Up:** Estimating individual WBS work items and summing them up. Highly accurate, but time-consuming and expensive.
- **Parametric Modeling:** Uses mathematical models based on project characteristics (e.g., $50 per line of code). Highly accurate if historical data is good. Often computerized.
    - **Computerized tools**: such as project management software and spreadsheet. Software tools simplify the estimation and facilitate rapid consideration of many cost estimate alternatives

Cost Estimation Outputs

- **Cost Estimate:** is a summary or detailed likely cost of the resources required to complete the project.
    - Includes all resources such as labor, material, supplies etc.
    - Expressed in unit of currency
- **Supporting details** includes WBS, Estimate basis , Assumption made, range of possible thresholds
- **Cost management Plan:** is part of overall project plan includes how cost variances will be managed e.g. formal or informal reporting, detailed or broad information.

Process 3: Determining the Budget

Allocating the overall cost estimate to individual work items over time.

- **Input:** The WBS is required because it defines the work items.
- **Output:** A **Cost Baseline** (a time-phased budget used to measure performance).

Example :

WBS	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Total
Labor	5000	5000	5000	5000	5000	5000	5000	5000	40000
HW	10000								10000
SW	2000								2000
Total									52000

Process 4: Controlling Costs

Monitoring performance, ensuring only appropriate changes hit the baseline, and informing stakeholders. **Earned Value Management (EVM)** is the primary technique used here.

7. Earned Value Management (EVM)

EVM integrates scope, time, and cost data to determine how well the project is meeting its goals.

EVM Core Terms

Term	Meaning	Definition
PV	Planned Value (Budget)	The approved budget planned to be spent on an activity by a given date.
AC	Actual Cost	The total direct and indirect costs actually spent so far.
EV	Earned Value	The value of the work *actually completed* (**$EV = PV \times RP$**).
RP	Rate of Performance	Ratio of actual work completed to work planned to be completed (e.g., planned to finish 100%, but only finished 50%, so RP = 50%).
BAC	Budget at Completion	The original planned total budget.
EAC	Estimate at Completion	What the *total* project is expected to cost now, based on current performance.
ETC	Estimate to Complete	Forecasted *time* to complete the job.
VAC	Variance at Completion	How much over/under budget we expect to be at the end.

EVM Formulas & Rules of Thumb

Rule of thumb: Negative numbers or ratios under 1.0 (100%) mean the project is doing poorly (over budget or behind schedule).

- **Cost Variance (CV):** $CV = EV - AC$ *(Negative = Over budget)*
- **Schedule Variance (SV):** $SV = EV - PV$ *(Negative = Behind schedule)*
- **Cost Performance Index (CPI):** $CPI = \frac{EV}{AC}$ *(< 1.0 = Over budget)*
- **Schedule Performance Index (SPI):** $SPI = \frac{EV}{PV}$ *(< 1.0 = Behind schedule)*
- **Estimate at Completion (EAC):** $EAC = \frac{BAC}{CPI}$
- **Estimate to Complete (ETC):** $ETC = \frac{Original\ Time}{SPI}$

EVM Exercise

Scenario: Install 10 printer drivers in 10 weeks. Budget: $100,000$(10,000 per printer). Plan: 1 printer per week.

Status at Week 5: Finished 4 printers. Actual cost so far: $47,500.

Step 1: Find the Base Numbers

- **PV:** At week 5, you planned to finish 5 printers (5 x $10,000) = $50,000
- **AC:** Given as $47,500
- **RP:** 4 finished / 5 planned = 0.8
- **EV:** $PV \times RP$ $(50,000 \times 0.8)$ = $40,000
- **BAC:** $100,000

Step 2: Calculate Performance

- **CV:** $40,000 - 47,500$ = **-$7,500** *(Over budget)*
- **SV:** $40,000 - 50,000$ = **-$10,000** *(Behind schedule)*
- **CPI:** $40,000/47,500$ = **0.84**
- **SPI:** $40,000/50,000$ = **0.80**
- **EAC (New Budget Forecast):** $100,000/0.84$ = **$119,000**
- **ETC (New Time Forecast):** 10 weeks / 0.80 = **12.5 weeks**

Summary

- Project cost management is a traditionally weak area of IT projects, and project managers must work to improve their ability to deliver projects within approved budgets
- Main processes include
    - Plan cost management
    - Estimate costs
    - Determine the budget
    - Control costs

اللهم اجعل هذا المسعى بداية لكل خير، ونهاية لكل تعب

ملاحظة ختامية

الحمد لله الذي بنعمته تتم الصالحات، والصلاة والسلام على معلم البشرية الخير، نبينا محمد وعلى آله وصحبه أجمعين.

تم بفضل الله وتوفيقه الانتهاء من نهاية الملخص

رسالة ختامية: كما ذكرتُ في المقدمة، فإن هذا الجهد أمانة بين يديك؛ فلا تجعل استخدامه يخرج عن إطار طلب العلم ، وابتعد عن كل ما يخدش ضمير الطالب المسلم من غش أو استغلال غير أخلاقي.

تذكير أخير: ما كان في هذا الملخص من صواب فمن الله وحده، وما كان فيه من خطأ أو نسيان فمني ومن الشيطان. وكما هو موضح في بيانات التواصل، بابي مفتوح لكل تصحيح أو إضافة تثري هذا العمل وتخدم الدفعات القادمة.

دعاء: اللهم انفعنا بما علمتنا، وعلمنا ما ينفعنا، وزدنا علماً. اللهم اجعل ما قرأناه حجة لنا لا علينا، ووفقنا جميعاً في اختباراتنا وفي حياتنا العملية لخدمة دينه ووطننا.

إعداد الطالب: إلياس حسان بابور

الموقع الإلكتروني: kau-one.web.app

رقم التواصل: 0530717788

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تم بحمد الله.